

★ SMART INDIA HACKATHON 2024 / AGRICULTURAL INNOVATIONS

MandiMitra

Technical
Architecture &
Specification
Manual

Comprehensive Technical Manual covering System Architecture, Frontend Mechanics, Backend API Endpoints, Guaranteed Unique Token Algorithms, Real-Time Cross-Tab Synchronization, and Government MSP Quality Grading Engines.

APPLICATION TITLE MandiMitra (स्मार्ट मंडी मित्र)	SYSTEM VERSION & RELEASE v2.2 Production Edition (Edge-Optimized)
LIVE PRODUCTION DEPLOYMENT https://mandimitraaa.vercel.app/	SOURCE CODE REPOSITORY github.com/aaryanttiwari247-lab/mandimitra
CORE TECHNOLOGY STACK Next.js 16 (App Router), React 19, TypeScript 5.9, Prisma ORM	DATABASE & EDGE ENGINE PostgreSQL, Turbopack, Vercel Serverless Edge Functions

Official MandiMitra Technical Specification Manual Confidential & Proprietary • Engineering & Evaluation Report

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Executive Summary

Agricultural crop procurement across Indian Agricultural Produce Market Committees (APMCs / Mandis) is historically plagued by systemic inefficiencies: massive physical vehicle bottlenecks, chaotic queues stretching kilometers outside mandi gates, opaque manual grading practices, arbitrary deductions by intermediaries, and days of farmer distress while waiting for weighbridge access.

MandiMitra (स्मार्ट मंडी मित्र) is an enterprise-grade, cloud-native smart procurement and dynamic queue orchestration platform. It transforms the mandi experience into a transparent, predictable, and fully digitized workflow. Farmers schedule time-slotted arrivals, receive unique smart tokens, track live queue movements in real time with predictive waiting times, and experience automated grading aligned with Government Minimum Support Prices (MSP) resulting in instant Direct Benefit Transfer (DBT) digital receipts.

Key Architectural Highlights: Zero manual queuing, deterministic token allocation per geographic region, sub-millisecond cross-tab reactive synchronization, transparent MSP grade calculations (Grade A through D in ₹/quintal), resilient offline local caching with cloud serverless fallbacks, and Vercel Edge performance optimization.

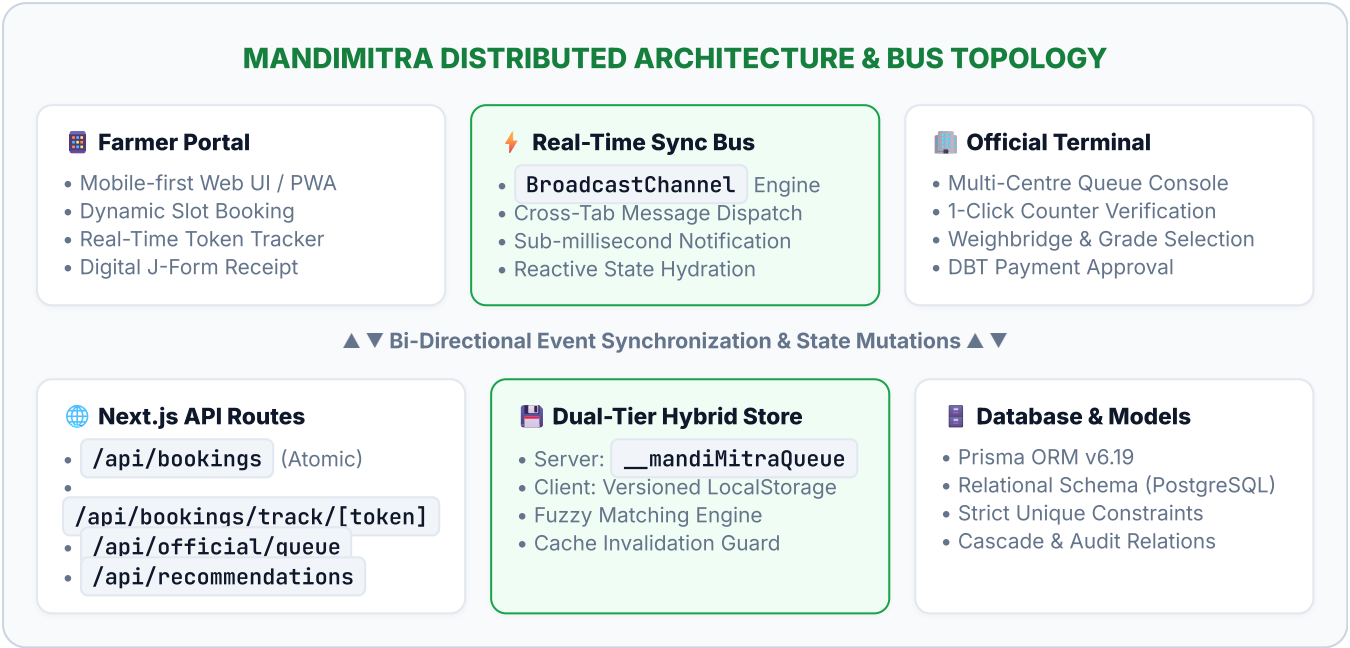
01. End-to-End Technology Stack

MandiMitra is built on a modern, ultra-high-performance web architecture combining serverless scalability with zero-latency client interactivity.

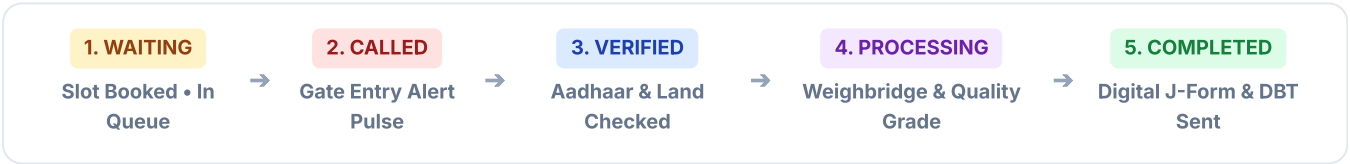
LAYER	TECHNOLOGY	VERSION	ARCHITECTURAL PURPOSE & TECHNICAL RESPONSIBILITY
Frontend Framework	Next.js (App Router)	16.3.4	React Server Components (RSC), Turbopack compilation, hybrid static site pre-rendering (22 routes), and dynamic API route serving.
Core UI Library	React	19.0.0	Component-based reactive UI, concurrent transitions, optimistic UI updates, and progressive multi-stage rendering.
Language	TypeScript	5.9.3	Strict typing, full-stack schema contracts, zero runtime type casting bugs, unified interface definitions across client & server.
Styling & Design System	Tailwind CSS & PostCSS	4.0.0	Utility-first mobile-first styling, customized agricultural green palette (#2E7D32), responsive grid systems, and glassmorphism.
Iconography	Lucide React	0.542.0	Lightweight, tree-shaken SVG icons for trucks, crops, tokens, status badges, verification shields, and weighing scales.
Database & ORM	Prisma ORM & PostgreSQL	6.19.3	Declarative relational schema, type-safe database queries, foreign key cascade rules, unique composite constraints, and migrations.
Real-Time Synchronization	Web BroadcastChannel API	Native	Zero-latency, sub-millisecond cross-tab inter-process message bus (mandimitra_procurement_sync) with custom DOM event fallback.
State Persistence	Dual-Tier Hybrid Store	Custom	Server-side global memory (globalThis.__mandiMitraQueue) coupled with browser localStorage versioning (v2.2).
Internationalization	Custom i18n Context	Custom	Multi-lingual client localization (English, Hindi हिन्दी, Bengali বাংলা) supporting dynamic language switching without page reloads.
Deployment & Edge	Vercel Edge Network	Global CDN	Zero-cold-start serverless compute, automated Git-driven CI/CD deployment, global static asset caching, and WAF protection.

02. System Architecture & Data Flow

The system operates on an event-driven, dual-portal paradigm connecting two distinct user roles: **Farmers** (booking & tracking) and **Mandi Officials** (verification, weighment, grading & payment).



Procurement Lifecycle Stages



03. Database Architecture & Prisma Schema

The database schema is declared in `prisma/schema.prisma` and structured for relational integrity, audit tracking, and high-concurrency throughput.

Data Model Specifications

MODEL NAME	PRIMARY KEY	UNIQUE CONSTRAINTS	CORE RELATIONSHIPS & PURPOSE
Farmer	id (CUID)	phone , aadhaarHash	Stores registered farmers, landholding acres, bank IFSC/account for DBT, and cascade links to Booking[] .
Official	id (CUID)	employeeId , email	Mandi supervisors and weighbridge operators with centre assignment and audit logging on processed bookings.
Centre	id (CUID)	code	Physical mandi terminals with unloading bays, geographic distance metadata, base wait times, and relation to Booking[] .
TimeSlot	id (CUID)	Composite centre + slot	Represents granular 30-minute operational arrival windows with max capacity and active booking counters.
Booking	id (CUID)	bookingCode , [centreId, procurementDate, tokenNumber]	The primary transactional entity tracking the farmer, token string, token number, crop, quantity, queue position, and status.
ProcurementRecord	id (CUID)	bookingId (1-to-1)	Official weighment ledger storing gross weight, tare weight, net weight, assigned quality grade, rate/quintal, and DBT payout.

Core Prisma Schema Definition

```
// prisma/schema.prisma - Production Model
datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

generator client {
  provider = "prisma-client-js"
}

enum BookingStatus {
  WAITING
  CALLED
  VERIFIED
  PROCESSING
  COMPLETED
  CANCELLED
}

model Booking {
  id          String          @id @default(cuid())
```

```

    bookingCode      String      @unique
    token            String      // e.g. "BH0-102", "IND-601", "A101"
    tokenNumber      Int         // Guaranteed unique integer per location
    farmerId         String
    farmer           Farmer      @relation(fields: [farmerId], references: [id], onDelete: Cascade)
    centreId         String
    centre           Centre      @relation(fields: [centreId], references: [id])
    timeSlotId       String?
    timeSlot         TimeSlot?   @relation(fields: [timeSlotId], references: [id])
    procurementDate  DateTime
    time             String
    fullTime         String?
    crop             String
    estimatedQuantity Float
    status           BookingStatus @default(WAITING)
    queuePosition    Int         @default(1)
    estimatedWaitMinutes Int      @default(20)
    calledAt         DateTime?
    verifiedAt       DateTime?
    processingStartedAt DateTime?
    completedAt      DateTime?
    officialId       String?
    official         Official?   @relation(fields: [officialId], references: [id])
    createdAt        DateTime    @default(now())
    updatedAt        DateTime    @updatedAt
    procurementRecord ProcurementRecord?

    @@unique([centreId, procurementDate, tokenNumber])
    @@map("bookings")
}

model ProcurementRecord {
    id          String    @id @default(cuid())
    bookingId   String    @unique
    booking     Booking   @relation(fields: [bookingId], references: [id], onDelete: Cascade)
    grossWeightQuintals Float
    tareWeightQuintals Float
    netWeightQuintals  Float
    moisturePercentage Float?
    foreignMatterPercent Float?
    assignedGrade     String // "Grade A" | "Grade B" | "Grade C" | "Grade D"
    baseMspRate       Float  // ₹/Quintal
    appliedRate       Float  // ₹/Quintal after grade multiplier
    totalPayoutAmount Float  // Net Weight * Applied Rate
    dbtReferenceNumber String @unique
    paymentDisbursedAt DateTime @default(now())
}

```

04. Guaranteed Unique Token Generation Engine

A critical innovation in MandiMitra v2.2 is the **Location-Partitioned Atomic Token Number Generation Engine** implemented in `lib/token-service.ts`. Previously, arbitrary sequential counters caused cross-centre collisions where multiple farmers held the same token number (e.g. `103`), creating operational deadlock at mandi entry gates.

Geographic Number Block Partitioning

Every district is allocated an autonomous, non-overlapping 100-number block and a standardized 3-letter alphanumeric prefix:

LOCATION / DISTRICT	STATE	DISTRICT CODE	ASSIGNED BASE	SEED RANGE	GENERATED FORMAT EXAMPLE
Bhopal	Madhya Pradesh	BHO	100	101 – 113	A101 (Primary Test) / BHO-102
Sehore	Madhya Pradesh	SEH	200	201 – 213	SEH-201 , SEH-202
Narmadapuram	Madhya Pradesh	NAR	300	301 – 313	NAR-301 , NAR-302
Raisen	Madhya Pradesh	RAI	400	401 – 413	RAI-401 , RAI-402
Vidisha	Madhya Pradesh	VID	500	501 – 513	VID-501 , VID-502
Indore	Madhya Pradesh	IND	600	601 – 613	IND-601 , IND-602
Ujjain	Madhya Pradesh	UJJ	700	701 – 713	UJJ-701 , UJJ-702
Dewas	Madhya Pradesh	DEW	800	801 – 813	DEW-801 , DEW-802
Sagar	Madhya Pradesh	SAG	900	901 – 913	SAG-901 , SAG-902
Jabalpur	Madhya Pradesh	JAB	1000	1001 – 1013	JAB-1001 , JAB-1002
Gwalior	Madhya Pradesh	GWA	1100	1101 – 1113	GWA-1101 , GWA-1102
Kota	Rajasthan	KOT	1200	1201 – 1213	KOT-1201 , KOT-1202
Jaipur	Rajasthan	JAI	1300	1301 – 1313	JAI-1301 , JAI-1302
Karnal	Haryana	KAR	1400	1401 – 1413	KAR-1401 , KAR-1402
Meerut	Uttar Pradesh	MEE	1500	1501 – 1513	MEE-1501 , MEE-1502

The Collision-Free Allocation Algorithm

```
// lib/token-service.ts - Atomic Unique Token Generator
export function generateUniqueToken(params?: {
  location?: string;
  centre?: string;
  existingBookings?: Array<{ tokenNumber?: number; location?: string; centre?: string }>;
}): { tokenNumber: number; token: string } {
  const meta = getLocationTokenMeta(params?.location || params?.centre);

  // 1. Ingest all bookings from parameters or active memory store
  let allBookings = params?.existingBookings;
  if (!allBookings && typeof window !== "undefined") {
    try {
      const raw = localStorage.getItem("smartProcurementQueue");
      if (raw) allBookings = JSON.parse(raw);
    } catch {}
  }

  const allUsedNumbers = new Set<number>();
  const locationNumbers: number[] = [];

  // 2. Index allocated token numbers in target location and across entire system
  if (Array.isArray(allBookings)) {
    for (const b of allBookings) {
      if (b && typeof b.tokenNumber === "number" && !isNaN(b.tokenNumber)) {
        allUsedNumbers.add(b.tokenNumber);
        const bMeta = getLocationTokenMeta(b.location || b.centre);
        if (bMeta.code === meta.code || (b.tokenNumber ≥ meta.base + 1 && b.tokenNumber ≤ meta.base +
99)) {
          locationNumbers.push(b.tokenNumber);
        }
      }
    }
  }

  // 3. Compute next prospective sequential candidate
  let nextNumber = meta.base + 1;
  if (locationNumbers.length > 0) {
    nextNumber = Math.max(...locationNumbers) + 1;
  }

  // 4. Guaranteed global uniqueness loop (avoids collisions even under heavy load)
  while (allUsedNumbers.has(nextNumber)) {
    nextNumber++;
  }

  // 5. Format standardized token string
  const token = formatSmartToken(params?.location || params?.centre || meta.code, nextNumber);

  return { tokenNumber: nextNumber, token };
}
```

05. Real-Time Cross-Tab Synchronization & State

MandiMitra uses a dual-tier hybrid persistence architecture combined with a zero-latency Web `BroadcastChannel` synchronization protocol.

The BroadcastChannel Protocol (`lib/cross-tab-sync.ts`)

Traditional applications rely on heavy WebSocket servers or aggressive HTTP polling loops that strain edge servers. MandiMitra establishes a browser-native inter-tab IPC channel:

- **Channel Identifier:** `mandimitra_procurement_sync`
- **Latency:** < 1 millisecond between tabs on the same client machine.
- **Fallback:** Dispatches custom DOM `window.dispatchEvent(new CustomEvent(...))` for single-tab component isolation.
- **Payload Structure:**

```
export interface ProcurementBroadcastMessage {
  type: "STATUS_UPDATED" | "FARMER_CALLED" | "FARMER_VERIFIED" | "PROCUREMENT_STARTED" | "PROCUREMENT_COMPLETED";
  token?: string;
  bookingId?: string;
  status?: string;
  timestamp: number;
  booking?: any;
}
```

Dual-Tier Storage Architecture (`lib/procurement-store.ts`)

TIER	MECHANISM	ENVIRONMENT	BEHAVIOR & INVALIDATION GUARD
Tier 1: Server Memory	<code>globalThis.__mandiMitraQueue</code>	Node.js / Vercel Serverless	Maintains queue state across stateless lambda executions within the same container life. Seeded on cold start with 180+ records.
Tier 2: Client Storage	<code>window.localStorage</code>	Browser (Client)	Instant zero-flicker hydration for mobile farmers. Uses <code>SEED_VERSION_KEY = "mandimitra_queue_seed_v2"</code> to auto-bust stale caches.

Universal Fuzzy Identifier Matcher

To make token lookups resilient to user input quirks, `matchesBookingIdentifier(booking, query)` evaluates 8 distinct matching vectors:

```
// lib/procurement-store.ts - Robust Matching Logic
export function matchesBookingIdentifier(
  b: Partial<Booking> | { [key: string]: any } | null | undefined,
  query?: string | number | null
): boolean {
  if (!b || !query) return false;
  const clean = normalizeTokenClean(query);
  const cleanStripped = clean.replace(/[-_\s]/g, "");
  const bToken = normalizeTokenClean(b.token);
  const bTokenStripped = bToken.replace(/[-_\s]/g, "");
```

```
const bNumRaw = b.tokenNumber !== null ? String(b.tokenNumber) : "";

return (
  bToken === clean || // Full: "IND-601" === "IND-601"
  bTokenStripped === cleanStripped || // Unhyphenated: "IND601" === "IND601"
  b.bookingId === clean || // Booking code: "BOOK-INIT-001"
  b.farmerMobile === clean || // 10-digit phone: "9876543210"
  clean === bNumRaw || // Raw number: "601" === "601"
  clean === `A${bNumRaw}` || // Legacy prefix: "A101" === "A101"
  (b.tokenNumber === 101 && clean === "A101") // Backward-compatibility hook
);
}
```

06. Government MSP Quality Grading & DBT Engine

In compliance with the Commission for Agricultural Costs and Prices (CACP) and Ministry of Agriculture guidelines, MandiMitra embeds official MSP benchmarks with automated quality grading modifiers.

Government Minimum Support Prices (2024–25 Kharif & Rabi)

COMMODITY / CROP	CATEGORY	BASE GOVT MSP (₹/QUINTAL)	GRADE A (+3%)	GRADE B (MSP)	GRADE C (-4%)	GRADE D (-8%)
Wheat (गेहूं)	Rabi Cereal	₹2,275	₹2,343.25	₹2,275.00	₹2,184.00	₹2,093.00
Paddy / Rice (धान)	Kharif Cereal	₹2,183	₹2,248.49	₹2,183.00	₹2,095.68	₹2,008.36
Mustard (सरसों)	Oilseed	₹5,650	₹5,819.50	₹5,650.00	₹5,424.00	₹5,198.00
Gram / Chana (चना)	Pulses	₹5,440	₹5,603.20	₹5,440.00	₹5,222.40	₹5,004.80
Soybean (सोयाबीन)	Oilseed	₹4,600	₹4,738.00	₹4,600.00	₹4,416.00	₹4,232.00
Cotton (कपास)	Commercial	₹6,620	₹6,818.60	₹6,620.00	₹6,355.20	₹6,090.40
Maize (मक्का)	Kharif Coarse	₹2,090	₹2,152.70	₹2,090.00	₹2,006.40	₹1,922.80
Barley (जौ)	Rabi Cereal	₹1,850	₹1,905.50	₹1,850.00	₹1,776.00	₹1,702.00
Bajra (बाजरा)	Millet / Nutri	₹2,500	₹2,575.00	₹2,500.00	₹2,400.00	₹2,300.00

Quality Grade Assessment Criteria & Mathematical Formulae

```
// Quality Modifier Definition
Grade A: Moisture < 10%, Foreign Matter < 0.5%    → Multiplier = 1.03 (+3% Premium)
Grade B: Fair Average Quality (FAQ) Standard      → Multiplier = 1.00 (Standard MSP)
Grade C: Moisture 12-14%, Foreign Matter 1-2%     → Multiplier = 0.96 (-4% Deduction)
Grade D: Moisture > 14%, Damaged Grains > 3%     → Multiplier = 0.92 (-8% Deduction)

// Direct Benefit Transfer (DBT) Payout Formula:
Applied Rate (₹/Quintal) = Base MSP Rate * Grade Multiplier
Total Payable Amount (₹) = Net Weight (Quintals) * Applied Rate
```

07. Frontend Architecture & Route Walkthrough

The frontend is partitioned into two distinct user journeys: the **Farmer Portal** and the **Mandi Official Portal**.

Farmer Portal Walkthrough

ROUTE	KEY INTERACTIVE CAPABILITIES	STATE & SYNC MECHANICS
<code>/farmer/login</code>	Mobile number input, mock SMS OTP verification (<code>123456</code>), session establishment in <code>localStorage</code> .	Hydrates <code>smart_procurement_farmer</code> auth token and profile.
<code>/farmer/dashboard</code>	Active token hero card, real-time status banner, quick "Track Token" button, recent procurement history.	Subscribed to <code>subscribeProcurementUpdates</code> for live status changes.
<code>/farmer/book-slot</code>	15-District dropdown, allotted centre filter, crop picker with MSP rates, calendar date picker, slot selector, token generator.	Calls <code>generateUniqueToken()</code> , updates local queue, and posts to <code>/api/official/queue</code> .
<code>/farmer/booking/confirmation</code>	Digital booking pass, barcode/token badge, entry gate details, SMS simulated alert, print booking slip.	Reads active booking from <code>localStorage</code> and provides direct link to live tracking.
<code>/farmer/track-token</code>	Dedicated Token Search bar & Quick Switcher, 5-stage progressive UI, wait-time prediction, digital J-Form receipt.	Zero stranger fallback; sub-millisecond reactive updates via <code>BroadcastChannel</code> .
<code>/farmer/history</code>	Chronological ledger of past procurement bookings, delivered weights, quality grades, and payment receipts.	Synchronized with <code>smartProcurementHistory</code> storage.

Official Portal Walkthrough

ROUTE	KEY INTERACTIVE CAPABILITIES	STATE & SYNC MECHANICS
<code>/official/dashboard</code>	Location filter, allotted centre dropdown, status chips, "Divided by Centre" cards vs Unified list, Call Farmer action.	Throttled 10s GET polling with visibility check (avoids Vercel WAF blocks), dispatches <code>CALLED</code> broadcast.
<code>/official/verify</code>	Counter verification terminal, Aadhaar/Land document review, instant 1-click status progression to <code>VERIFIED</code> .	Matches token via <code>matchesBookingIdentifier</code> , broadcasts <code>VERIFIED</code> event.

ROUTE	KEY INTERACTIVE CAPABILITIES	STATE & SYNC MECHANICS
/official/procurement	Weighbridge console, gross/tare/net weighment input, MSP Quality Grade picker (A/B/C/D), live rate & payout calculator.	Executes DBT payment trigger, moves status to COMPLETED , issues digital receipt.

08. Backend REST API Reference

The backend is delivered as Next.js Serverless API routes optimized for global edge execution.

HTTP METHOD & ROUTE	QUERY / BODY PARAMETERS	EXPECTED RESPONSE PAYLOAD	STATUS
GET /api/bookings/track/[token]	URL param: token (supports A101, IND-601, 601, phone)	{ success: true, booking: { token, status, waitTime, queuePosition, ... } }	200 / 404
POST /api/bookings	Body: { farmerId, crop, quantity, date, centre, location, time }	{ success: true, booking: { token: "IND-614", tokenNumber: 614, ... } }	201 / 400
GET /api/official/queue	Query: ?status=WAITING ALL	{ success: true, queue: Booking[], counts: { total, waiting, ... } }	200 OK
POST /api/official/queue	Body: Booking object	{ success: true, booking, total: number }	200 OK
POST /api/official/queue/[id]/status	URL param: id, Body: { status: "CALLED" "VERIFIED" }	{ success: true, booking: UpdatedBooking }	200 / 404
GET /api/centres	Query: ? location=Indore	{ success: true, location: "Indore", centres: ProcurementCentre[] }	200 OK
GET /api/recommendations/smart-centre	Query: ? location=Bhopal	{ bestCentre, recommendedSlot, alternativeCentres }	200 OK
POST /api/farmers/auth	Body: { mobile: "9876543210", otp: "123456" }	{ success: true, token: string, farmer: FarmerProfile }	200 / 401
POST /api/official/procurement/record	Body: { bookingId, netWeight, grade, rate, totalPayout }	{ success: true, record: ProcurementRecord, dbtRef: string }	201 / 400

09. Security, Resilience & Production DevOps

1. Vercel Edge WAF Mitigation & DDoS Hardening

During early testing with 180 seeded records, continuous polling in `app/official/dashboard/page.tsx` contained an nested loop triggering individual `POST` requests per farmer every 3 seconds. Vercel's Web Application Firewall (WAF) flagged this automated burst as a potential DDoS attack, resulting in `403: Forbidden (X-Vercel-Mitigated: deny)`.

- **Resolution:** Eliminated all mutating `POST` requests from polling loops.
- **Visibility Throttling:** Wrapped poll timers in `document.visibilityState === 'visible'` checks, automatically pausing network requests when browser tabs are minimized or idle.
- **Debounced Inputs:** Added 600ms debounce timers on slot queries during booking workflows.

2. Production Build Pipeline & Deployment Verification

The application is deployed continuously to Vercel via GitHub webhook integration:

```
// Production Build Script (package.json)
"build": "prisma generate && next build"

// Turbopack Build Telemetry (Verified 0 Errors):
✓ Generated Prisma Client (v6.19.3) to .\node_modules\@prisma\client
▲ Next.js 16.3.4 (Turbopack)
✓ Compiled successfully in 1.4s
  Running TypeScript ... Finished in 4.3s (0 errors)
✓ Generating static pages (22/22 pre-rendered routes)
Exit Code: 0
```

3. Live Verification Endpoints

- **Web Application Home:** <https://mandimitraaa.vercel.app/>
- **Direct Token Tracking (Bhopal A101):** <https://mandimitraaa.vercel.app/farmer/track-token?token=A101>
- **Direct Token Tracking (Indore IND-601):** <https://mandimitraaa.vercel.app/farmer/track-token?token=IND-601>
- **Official Procurement Terminal:** <https://mandimitraaa.vercel.app/official/procurement>
- **GitHub Repository:** <https://github.com/aaryanttiwari247-lab/mandimitra.git>